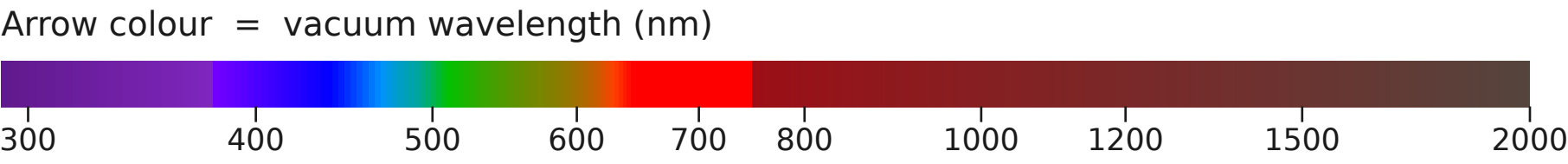


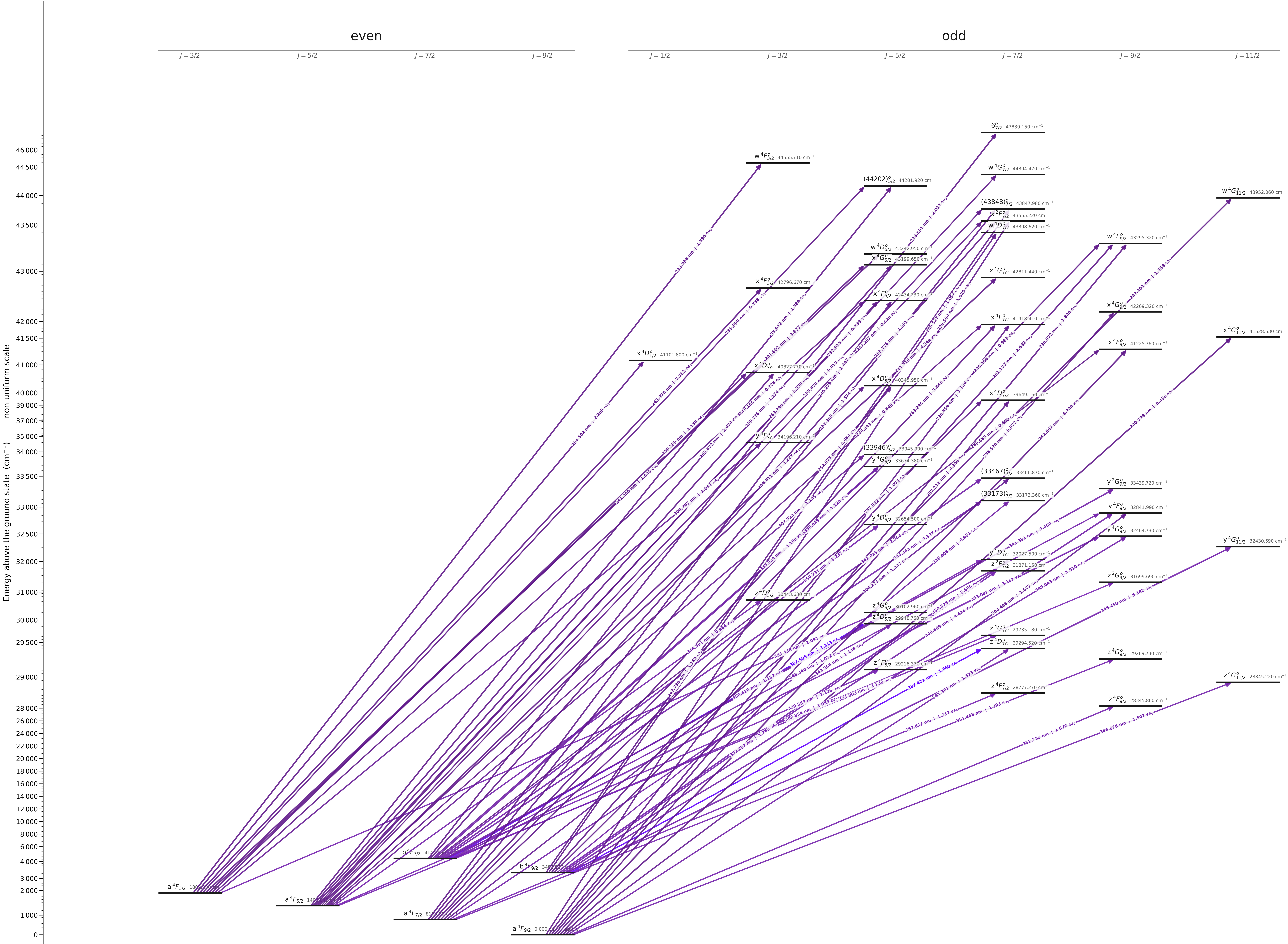
Cobalt

Arrow label: vacuum wavelength | reduced dipole matrix element $[U||er||J']$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).



Co (Co I). Strongest classified transitions below 2 μm (NIST): 51 levels, 70 lines.

Co⁺ ionisation limit 63564.60 cm⁻¹ = 7.88101 eV (157.320 nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	[U er J'] (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
b ⁴ F _{7/2} - 6 ⁵ ₂	228.8513	228.7808	1309.9878	2.017	8.600e+07	-	-	
a ⁴ F _{9/2} - x ⁴ G ^o _{11/2}	240.7983	240.7251	1244.9940	5.456	3.600e+08	-	-	
a ⁴ F _{5/2} - x ⁴ G ^o _{9/2}	241.5191	241.4457	1241.2787	4.349	3.400e+08	-	-	
a ⁴ F _{3/2} - x ⁴ G ^o _{5/2}	241.6024	241.5290	1240.8506	3.877	3.600e+08	-	-	
a ⁴ F _{9/2} - x ⁴ F ^o _{8/2}	242.5668	242.4931	1235.9172	4.748	3.200e+08	-	-	
a ⁴ F _{7/2} - x ⁴ F ^o _{7/2}	243.2947	243.2209	1232.2193	3.845	2.600e+08	-	-	
a ⁴ F _{5/2} - x ⁴ F ^o _{5/2}	243.7396	243.6657	1229.9702	3.339	2.600e+08	-	-	
a ⁴ F _{3/2} - x ⁴ F ^o _{3/2}	243.9778	243.9038	1228.7695	2.782	2.700e+08	-	-	
b ⁴ F _{9/2} - w ⁴ F ^o _{8/2}	251.1774	251.1018	1193.5487	2.682	9.200e+07	-	-	
a ⁴ F _{9/2} - x ⁴ D ^o _{7/2}	252.2122	252.1363	1188.6519	4.359	3.000e+08	-	-	
a ⁴ F _{7/2} - x ⁴ D ^o _{5/2}	252.9727	252.8967	1185.0781	3.664	2.800e+08	-	-	
a ⁴ F _{5/2} - x ⁴ D ^o _{3/2}	253.6724	253.5961	1181.8097	2.474	1.900e+08	-	-	
a ⁴ F _{3/2} - x ⁴ D ^o _{1/2}	254.5017	254.4253	1177.9586	2.209	3.000e+08	-	-	
b ⁴ F _{9/2} - y ⁴ F ^o _{8/2}	340.6091	340.5114	880.1658	4.416	1.000e+08	-	-	
b ⁴ F _{9/2} - y ⁴ G ^o _{11/2}	345.4498	345.3508	867.8323	5.182	1.100e+08	-	-	
b ⁴ F _{7/2} - y ⁴ D ^o _{5/2}	350.7315	350.6312	854.7635	3.237	8.200e+07	-	-	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).